Seminars in Arthritis and Rheumatism 50 (2020) 627–635

Contents lists available at ScienceDirect

Seminars in Arthritis and Rheumatism

journal homepage: www.elsevier.com/locate/semarthrit

The altered metabolism profile in pathogenesis of idiopathic inflammatory myopathies

Di Liu, Xiaoxia Zuo, Hui Luo, Honglin Zhu*

Department of Rheumatology, Xiangya Hospital, Central South University, 87 Xiangya Road, Changsha, Hunan 410008, People's Republic of China

ARTICLE INFO

Keywords:
IIM
Metabolism
Pathogenesis

ABSTRACT

Idiopathic inflammatory myopathies (IIMs) are a group of heterogeneous autoimmune diseases characterized by muscle weakness, muscle inflammation and extramuscular manifestations. Despite extensive efforts, the mechanisms of IIMs remain largely unknown, and treatment is still a challenge for physicians. Metabolism changes have emerged as a crucial player in autoimmune diseases, such as systemic lupus erythematosus (SLE) and rheumatoid arthritis (RA). However, little is known about metabolism changes in IIMs. In this review, we focus on the alteration of metabolism profile in IIMs, and the relationships with clinical information. We highlight the potential roles of metabolism in the pathogenesis of IIMs and discuss future perspectives for metabolic checkpoint-based therapeutic interventions.

© 2020 Elsevier Inc. All rights reserved.

Introduction

Idiopathic inflammatory myopathies (IIMs) are a heterogeneous group of systemic autoimmune diseases characterized by muscle weakness, muscle inflammation and extramuscular manifestations [1,2]. IIMs mainly include dermatomyositis (DM), polymyositis (PM), inclusion body myositis (IBM) and immune-mediated necrotizing myopathy (IMNM) [3]. The prevalence rate of IIMs varies greatly across studies, and the mortality rate is approximately 6% to 60% [4–6]. IIMs complicated with rapidly progressive interstitial lung disease (ILD) have the highest mortality rates, which range from 70% to 90% [7]. Histopathologically, DM is characterized by invasion of perivascular and perimysial areas by B lymphocytes, CD4+ T cells, macrophages and plasmacytoid dendritic cells (pDCs) [8]. CD8+ T cell infiltration and abnormal expression of major histocompatibility complex-I (MHC-I) on non-necrotic fibres are features of PM and IBM. Another feature of IBM is the appearance of rimmed vacuoles containing basophilic granules and β-amyloid in some fibres [9]. IMNM is different from other types of IIMs, as it is characterized by macrophage infiltration rather than by lymphocyte infiltration [10].

The mechanisms of IIMs remain largely unknown. Studies in the past decade have shown that genetic and environmental risk factors determine susceptibility to IIMs, and both immune and non-immune mechanisms contribute to muscle weakness and damage [1,11]. Among immune mechanisms, lymphocyte-mediated cytotoxicity, autoantibody activity, overexpression of MHC-I on muscle sarcolemma, and cytokine and chemokine signaling are the main innate and adaptive immune mechanisms [12]. In the pathogenesis of IIMs, CD4+ T cells infiltrating muscle tissue can differentiate into T helper 1 (Th1) or Th17 cells, produce the inflammatory factors interferon (IFN)-γ and interleukin-17 (IL-17), and ultimately prompt the upregulation of MHC-I in muscle fibres. CD8+ T cells, CD4+ CD28-null T cells and CD8+ CD28-null T cells have been shown to exert myotoxicity to muscle cells through perforin and granzyme B pathway and human leukocyte antigen (HLA) regulatory mechanisms [13,14]. In addition, Th1 cells can prompt B cells to differentiate and produce antibodies that are associated with distinct genotypes, clinical phenotypes and the IFN pathway [15]. Antibodies and complement system are deposited on endomysial capillary endothelial cells, further

Abbreviations: AA, arachidonic (20:4, n-6) acid; AGL, amylo-1,6-glucosidase, 4-alpha-glucanotransferase; ALDOA, aldolase; ATP, adenosine triphosphate; Breg, regulatory B cell; CADM, amyopathic dermatomyositis; CS, citrate synthase; COX, cyclooxygenase; DM, dermatomyositis; ENO3, enolase 3; ER, endoplasmic reticulum; FA, fatty acid; FBP2, fructose-1,6-bisphosphatase 2; GAPDH, glyceraldehyde 3-phosphate dehydrogenase; GC, germinal center; HIF-1, hypoxia-inducible factor-1; HLA, human leukocyte antigen; HMGCR, 3-hydroxy-3-methylglutaryl-CoA reductase; Hsp60, heat shock protein 60; IBM, inclusion body myositis; IF1, Inhibitory Factor 1; IFN, interferon; IIMs, idiopathic inflammatory myopathies; IL, interleukin; LTB4, leukotriene B4; ILD, interstitial lung disease; IMNM, immune-mediated necrotizing myopathy; LDHA, lactate dehydrogenase A; LPS, lipopolysaccharide; LTB4, Leukotriene B4; MD, mitochondrial disorders; MDH, malate dehydrogenase; MHC, major histocompatibility complex; MSAs, myositis-specific autoantibodies; mtDNA, mitochondrial DNA; NO, nitric oxide; OXPHOS, oxidative phosphorylation; PA, palmitic acid; PBMCs, peripheral blood mononuclear cells; PC, phosphatidylcholine; pDC, plasmacytoid dendritic cells; PDK4, pyruvate dehydrogenase kinase 4; PEKM, phosphofructokinase mutase; PET/CT, positron emission tomography and computed tomography; PGAM2, phosphoglycerate mutase 2; PKM2, pyruvate kinase isoform M2; PM, polymyositis; PPP, pentose phosphate pathway; RA, rheumatoid arthritis; SLE, systemic lupus erythematosus; SSc, systemic sclerosis; ROS, reactive oxygen species; TCA, tricarboxylic acid; Tfh, T follicular helper; Th, T helper; TLR, toll-like receptor; Tregs, regulatory T cells; TG, triacylglycerol; 2-DG, 2-deoxy-D-glucose; 18F-FDG, 18-fluorodeoxyglucose.

* Corresponding author.
E-mail address: Honglinzhu@csu.edu.cn (H. Zhu).

https://doi.org/10.1016/j.semarthrit.2020.05.008
0049-0172/© 2020 Elsevier Inc. All rights reserved.

activating a complement cascade reaction to induce muscle damage. In addition to immune mechanisms, non-immune mechanisms, such as endoplasmic reticulum (ER) stress, autophagy, and energy metabolism, have also been found to induce muscle damage [16]. Recently, increasing evidence has suggested that mitochondrial abnormalities and metabolic disturbances are present in IIM patients. Endurance exercise in DM/PM patients prompts a metabolic switch to oxidative phosphorylation (OXPHOS), simultaneously suppresses the inflammatory response in muscle tissues [17,18]. Altered muscle anabolism, catabolism and contraction can lead to irreversible structural damage of muscle fibers [1,19].

Metabolism is at the center of all biological processes. Distinct metabolic pathways are needed to satisfy specific functions [20]. Glycolysis, the tricarboxylic acid (TCA) cycle, fatty acid (FA) oxidation, FA synthesis, amino acid metabolism and the pentose phosphate pathway (PPP) are six major metabolic pathways [21,22]. Cellular metabolism not only produces energy for cells but also supports cell growth, differentiation and function. The emerging field of immunometabolism focuses on the functional links between intracellular metabolic pathways and the immune system [23]. Metabolic reprogramming in immune cells influences effector functions, such as differentiation, proliferation, migration, secretion of pro-inflammatory factors and anti-infection processes [24–26]. Recently, immunometabolism has been widely studied in the context of rheumatologic diseases. In systemic lupus erythematosus (SLE), CD4+ T cells display increases in glucose metabolism and defects in lipid metabolism [27]. In rheumatoid arthritis (RA), naive CD4+ T cells show dampened glycolysis, an enhanced PPP and low levels of adenosine triphosphate (ATP), and fibroblast-like synoviocytes exhibit a shift from OXPHOS to glycolysis. Inhibition of glycolysis has been found to significantly reduce arthritis in an animal model of RA [28,29]. In systemic sclerosis (SSc) and fibrotic disease, metabolism plays an important role in fibroblast and immune cell activation and reactive oxygen species (ROS) production [21]. Skeletal muscle, the main site for glucose disposal and amino acid storage, is the major regulator of metabolism [30]. However, little is known about the role of metabolism in IIMs. Here, we illustrate the alterations of metabolism profile in IIMs, explore their roles and discuss approaches to target metabolites or metabolic pathways for the treatment of IIMs.

Altered metabolism in IIMs

Glycolysis

18-fluorodeoxyglucose (18F-FDG), a kind of glucose analogue tracer, can be used to measure glucose uptake. In DM/PM patients, 18F-FDG positron emission tomography and computed tomography (PET/CT) scans displayed increased 18F-FDG uptake in muscular tissue, which has been found to be positively correlated with serum creatine (Cr) kinase levels and negatively correlated with muscle strength. In addition, uptake of 18F-FDG is higher in patients with PM and DM than in patients with clinical amyopathic dermatomyositis (CADM). DM/PM patients with rapidly progressive interstitial lung disease (ILD) also demonstrate enhanced 18F-FDG accumulation in lung lesions compared with patients with chronic ILD [31]. These results suggest that DM/PM patients exhibit increased glucose uptake in muscle tissues. A cohort of thirty-two muscle biopsies from DM, PM and sporadic IBM patients and controls was analyzed the proteins expression of energy metabolism using reverse phase protein microarrays. Glycolytic pyruvate kinase isoform M2 (PKM2), mitochondrial ATPase Inhibitory Factor 1 (IF1) and heat shock protein 60 (Hsp60) were upregulated in the muscle tissues of DM patients compared with other IIMs subtypes and controls, and PKM2 and IF1 may be specific biomarkers of DM [32]. In IBM, glycolytic enzymes, including glyceraldehyde 3-phosphate dehydrogenase (GAPDH), lactate dehydrogenase A (LDHA), amylo-1,6-glucosidase, 4-alpha-glucanotransferase (AGL), phosphofructokinase mutase (PFKM), fructose-1,6-bisphosphatase 2 (FBP2), and phosphoglycerate mutase 2 (PGAM2), were significantly reduced in the muscle tissues, which suggested repression of glycolytic metabolism. Loss of glycolytic enzymes, particularly glycogen debranching enzyme AGL, was obvious in the atrophic muscle of IBM, accompanied by preferential loss of fast-twitch structural proteins [32,33]. But whether it can contribute to muscle weakness in IBM is still unclear. The metabolic proteins cannot distinguish PM biopsies from controls, which is contrary to our previous study. In our previous work, global protein expression in muscle tissues of DM/PM patients suggested that proteins participating in glycolysis/gluconeogenesis and the hypoxia-inducible factor-1 (HIF-1) signaling pathway were upregulated [34]. We speculate that the reason for the discordant results in PM may be the different diagnostic criteria or the difference in disease durations, medications between the enrolled PM patients from the two studies. However, the patient information is not clear in the first study, so we can't compare with each other.

In the autoimmune inflammatory myositis mouse model, proteomic analyses of quadriceps muscles revealed that glycolysis, OXPHOS, cytoskeletal protein, ubiquitin proteasome and ER stress pathways were significantly altered [35]. Glycolysis and OXPHOS pathways were downregulated prior to muscular infiltration of inflammatory cells in the myositis model. In addition, glycolytic enzymes, including LDHA, PFKM, PGAM2, enolase 3 (ENO3), aldolase (ALDOA), and PKM2, were also decreased at the transcriptional level [36]. In C2C12 myoblasts, lipopolysaccharide (LPS) stimulation increased the expression of pyruvate dehydrogenase kinase 4 (PDK4) and altered glucose metabolism via the Jun N-Terminal Kinase pathway [36].

All of the above evidence indicates that altered glycolysis might exist in IIMs, varying by subtypes. The level of glycolysis significantly increases in DM, and decreases in IBM, and reveals no difference or increases in PM compared with controls. In the autoimmune inflammatory myositis mouse model, glycolysis is downregulated. The altered glycolysis may provide new biomarkers or therapeutic targets for IIMs, especially for IBM.

FA metabolism

Statins can lower cholesterol levels by inhibiting the key enzyme 3-hydroxy-3-methylglutaryl-CoA reductase (HMGCR) in the cholesterol synthesis pathway. Approximately 2–3/100,000 statin-treated persons per year develop IMNM in association with antibodies directed against HMGCR [37,38]. Many theories have been proposed to explain the mechanism of statin myopathy, including mitochondrial dysfunction and subsequent abnormal muscle energy metabolism [39]. The phenomenon indicates that there may be a link between lipid metabolism and autoimmune myopathies. IIM patients are at risk of developing metabolic syndrome [40,41]. Decreased high-density lipoprotein cholesterol (HDL-C) and increased triglyceride levels are common in the serum of DM/PM patients. In addition, serum levels of C-reactive protein are negatively correlated with HDL-C levels in DM [42,43]. To identify serum lipid profile alterations, a targeted lipidomic analysis was performed on DM and PM patients. Compared with that of healthy controls, the serum of DM/PM patients exhibited alterations in FA species, such as high levels of palmitic (16:0) acid (PA) and decreased arachidonic (20:4, n-6) acid (AA) [44]. PA is the most abundant systemic saturated FA (SFA) and is correlated with atrophy of myotubes in vitro. PA levels were also increased in peripheral blood mononuclear cells (PBMCs) of PM patients [45]. The reduced AA levels are most likely caused by the conversion of AA to prostaglandin and leukotriene subfamilies, in line with the previous researches of the increased expression of enzymes in cyclooxygenase (COX) and leukotriene B4 (LTB4) pathways and increased synthesis of prostaglandins and leukotrienes in muscles from DM/PM patients [46–48]. Prostaglandins and leukotrienes are powerful mediators of muscle inflammation and play

critical roles in muscle atrophy, impairment, and pain [46,47]. While some prostaglandins and leukotrienes have an effect on muscle repair. So targeting specific molecular pathways to suppress inflammation and avoid the effect on muscle repair may be an attractive therapy for IIMs. Furthermore the levels of three phosphatidylcholine (PC) and two triacylglycerols (TG) species containing polyunsaturated FA (PUFA) such as PC(36:4), PC(38:5),PC(38:4), TG(48:5) and TG (54:4) were lower and the levels of four TG species containing SFA and monounsaturated FA (MUFA) such as TG(46:1),TG(48:2), TG (48:1) and TG(50:1) were higher in the serum of DM/PM patients. The ratio imbalance of SFAs and PUFAs may also have an effect on muscle inflammation and performance [49,50]. The levels of several other lipid species were also correlated with muscle performance and serum levels of Cr kinase. Moreover, lipid profiles in serum and muscle tissue were altered after immunosuppressive treatment in IIM patients, especially the significant increased level of eicosadienoic (20:2, n-6) acid (EDA) and the eicosapentaenoic (20:5, n-3) acid (EPA) [44,51]. EDA can exert anti-inflammatory function by prostaglandin E2 and nitric oxide pathways [52]. Early in the 1970s, intramyocellular lipid droplets were found in PM patient muscle tissue [53]. Similarly, free (non-esterified) cholesterol was found to accumulate in the muscle tissue of patients with IBM [54].Lipodystrophy was found in patients with juvenile DM (JDM), which was associated with loss or redistribution of subcutaneous fat [55]. Dietary supplementation with PUFA improves skeletal muscle mass and strength, suggesting a role for FA in the treatment of muscle weakness associated with IIMs [56,57].

TCA cycle and OXPHOS

Mitochondria are the site of TCA cycle and OXPHOS, can regulate multiple signaling pathways by the production of ATP, nicotinamide adenine dinucleotide (NADH) and ROS. Among IIMs, mitochondrial dysfunction, especially cytochrome c oxidase-deficient muscle fibers and large-scale mitochondrial DNA (mtDNA) deletions are more prominent in IBM [58,59]. Mitochondrial abnormalities also exist in a subset of PM with mitochondrial pathology [60]. Proteomic analyses of muscle tissue of PM have shown deficiencies in respiratory complexes and TCA cycle enzymes. Biochemical experiments have further confirmed significant losses of the activity of citrate synthase (CS) and malate dehydrogenase (MDH). Enzyme histochemistry has revealed increases in succinate dehydrogenase (SDH) and loss of cytochrome c oxidase [61]. Transcript levels of genes encoding proteins related to mitochondrial activities are decreased in muscle biopsies from DM patients, translating into decreased electron transport chain complexes, OXPHOS and increased ROS [62]. More interestingly, abnormal SDH and cytochrome c oxidase activities in DM muscle are more pronounced in the atrophic perifascicular fibers [63]. The skeletal muscles of amyloid precursor protein-overexpressing (MCK-βAPP) mice have myopathic features of IBM. ^{1}H-decoupled^{13}C NMR spectroscopy of MCK-βAPP mouse muscle has revealed significant reductions in the levels of succinate and glutamate, which are the intermediates or by-products of the TCA cycle. These results are consistent with increased levels of lactate and further indicate reduced activity of the TCA cycle and a switch in energy metabolism from aerobic to anaerobic glycolysis in this animal model of IBM [64]. Mitochondrial abnormalities in myofibers can occur in DM, PM and IBM, and usually predict a poor response to corticosteroids, immunosuppressive therapy [65]. Although the exact machnism is unclear, the interaction between ER stress, autophagy, ROS and mitochondria may explain part of the process [60,62,66].

Amino acid metabolism

A targeted metabolomics analysis of serum/plasma and muscle metabolites in IBM, mitochondrial disorders (MD), non-mitochondrial neuromuscular disease (NMD) and controls revealed that amino acid biosynthesis pathways, such as the aspartate, glutamate, alanine and transsulfuration pathway, are dysregulated in IBM compared with controls. And the alternative NAD+ synthesis pathway, nucleotide synthesis precursors, and the Cr/creatinine (Cn) ratio are highly increased in IBM. Furthermore, IBM metabolomes clustered together with MD, suggesting an important contribution of mitochondrial abnormalities in the metabolic changes in IBM. The disease specific metabolomic fingerprints can also help diagnosis and treatment [67]. In the urine of patients with DM/PM, the levels of Cr, taurine, glycine, citrate, and choline-containing compounds are elevated [68]. In JDM, elevated urinary Cr/Cn ratios are positively correlated with disease damage. All these findings indicate that Cr pools are depleted in individuals with IIMs [69]. Cr plays a protective role against muscle atrophy and exerts antioxidant activity during C2C12 cell differentiation [70]. IIM patients may benefit from oral Cr supplements combined with exercise [71] (Fig. 1) (Table 1).

Metabolic reprogramming in immune cells in IIMs

Dynamic changes in cellular metabolism regulate immune cell growth, proliferation, activation and homeostasis restoration [20,72]. Both innate and adaptive immune cells are involved in muscle weakness and damage in IIMs [1]. Therefore, metabolic reprogramming of immune cells might play an important role in IIMs pathogenesis.

T cells

In peripheral blood, total T lymphocyte counts are significantly lower in active untreated IIM patients compared with inactive IIM patients and healthy controls. Moreover, lymphopenia is negatively correlated with disease activity and is reversed after glucocorticoid treatment [73–75]. Peripheral T cell subsets from IIM patients are characterized by increased populations of proinflammatory T cells, such as T follicular helper (Tfh) cells and Th17 cells; expansion of pro-apoptotic CD28-null cells; and decreased suppressor populations of regulatory T cells (Tregs) [74–76]. In the muscle tissue of patients with IIMs, diverse T cell subsets infiltrate and play distinct roles. Cytotoxic CD8+ effector T cells expressing granzyme B and perforin infiltrate non-necrotic muscle tissue in PM and IBM, which is a pathological feature of these conditions [13,77]. CD28-null T cells (CD4+ CD28-null and CD8+ CD28-null T cells) have been shown to be dominant among muscle-infiltrating T cells in DM/PM and form a subpopulation of cytotoxic, pro-inflammatory, apoptosis-resistant T cells [78,79]. Aside from pro-inflammatory and cytotoxic T cells, Tregs in inflamed muscle are close to effector cells, but the high proportions of Tregs coexisting with muscle inflammation suggest that their anti-inflammatory effects may be impaired in IIMs [80].

Metabolic reprogramming is closely linked with T cell activation, differentiation and function. Naive T cells and memory T cells are quiescent, have a low energy demand, and mainly perform OXPHOS and FA oxidation for energy [81]. Upon activation, effector T cells exhibit high rates of anabolic metabolism to support their effector functions. They adopt glycolysis and the TCA cycle to supply the energy for de novo protein, nucleic acid and lipid synthesis [82,83]. Manipulation of the metabolic program drives the differentiation of T cells into different effector or regulatory T cell subtypes [84]. Th1, Th2 and Th17 cells mainly rely on glycolysis to support their effector functions, but Tregs do not [85]. Instead, Tregs rely on increased levels of lipid oxidation [84]. It has been shown that inhibition of de novo FA synthesis suppresses the glycolytic-lipogenic pathway, which restrains the formation of Th17 cells and promotes the development of Tregs. Therefore, it is suggested that the activation of CD4+ and CD8+ effector T cells in IIMs may reprogram the

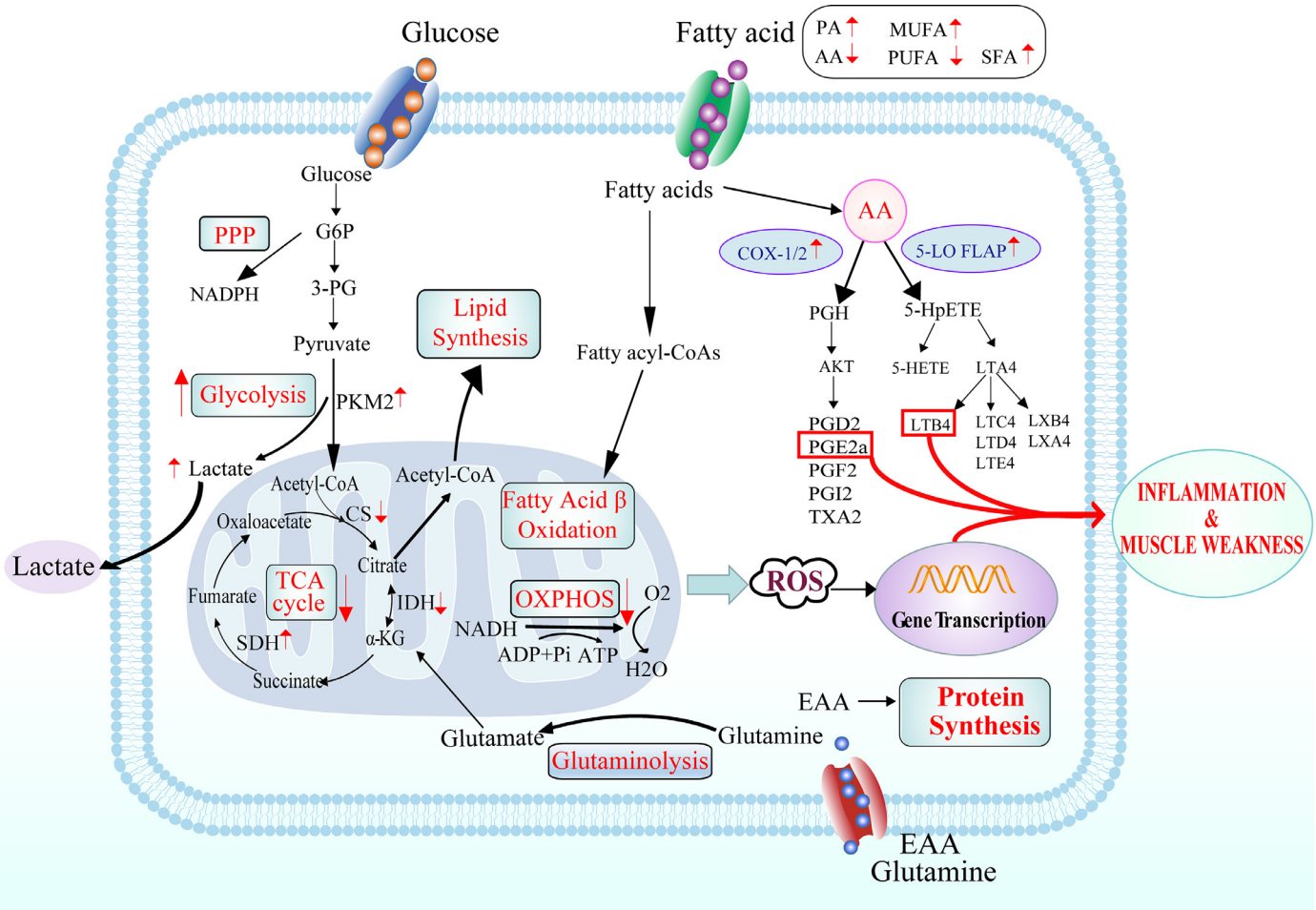

Fig. 1. The dysregulated metabolic pathways in DM and PM. The glycolysis level is increased in muscle tissues from DM patients. Mitochondrial abnormalities are associated with the inhibition of TCA cycle and OXPHOS and excessive production of ROS, finally lead to oxidative damage, muscle inflammation and weakness in DM/PM. The increased PA and the ratio imbalance of saturated SFA and PUFA in serum have an effect on muscle atrophy and inflammation in DM/PM. The AA in serum may get into skeletal muscle cells, converts into prostaglandin and leukotriene subfamilies and plays critical roles in muscle atrophy, impairment, and pain. PKM2, pyruvate kinase M 2; PPP, pentose phosphate pathway; CS, citrate synthase; SDH, succinate dehydrogenase; IDH, isocitrate dehydrogenase; α-KG, α-ketoglutarate; PA, palmitic acid; AA, arachidonic (20:4, n-6) acid; PUFA, polyunsaturated fatty acid; MUFA, monounsaturated fatty acid; SFA, saturated fatty acid; EAAs, essential amino acids; COX, cyclooxygenase; PGH, prostaglandin H; LTA4, leukotriene A4; TXA2, thromboxane A2; 5-LO, 5-lipoxygenase; FLAP, 5-LO activated protein; 5-HPETE, 5-hydroperoxyeicosatetraenoic acid.

metabolism of these cells to enable them to exert their proinflammatory or cytotoxic effects.

B cells

The role of B cells in the pathogenesis of IIMs has been suggested not only by the fact that activated B cells are increased in the peripheral blood of individuals with active DM/PM but also by the presence of myositis-specific autoantibodies (MSAs), and by the finding of B cells and plasma cells in muscle tissue [15,86,87]. Regulatory B cells (Bregs) have suppressive functions. CD19+ CD24-high CD38-high Bregs are expanded in JDM and are correlated with disease activity, while they are significantly decreased in peripheral blood in DM [88,89]. Autoantibodies are common in IIMs, and specific MSAs are associated with muscle pathology and clinical phenotypes [90–93]. The serum levels of B cell-activating factor are higher in patients with anti-Jo-1 autoantibodies and DM than in patients with other subtypes of IIMs [15,94,95]. B cells dominate in perivascular regions in DM, while plasma cells, a type of terminally differentiated B cells expressing CD138, dominate in muscle in IBM and PM, with numbers several times higher than those of B cells [86,96].

The metabolic needs of B cells differ depending on cell state, yet it is still unclear how metabolism directs the fate of B cells [97]. Upon viral stimulation, activated B cells show increased glycolytic activity [98–100]. Pre-B cells in early development and germinal center (GC) B cells in antigen-driven responses are the two main subsets of proliferative B cells. It has been shown that in bone marrow, compared with pro-B and small pre-B cells, large pre-B cells exhibit higher glucose uptake and ROS production [101]. In addition, GC B cells exhibit elevated glucose uptake and mitochondrial biogenesis [97]. These findings suggest that glucose metabolism is crucial for proliferative B cells to satisfy their high energy demand. However, for naive and memory B cells in blood and immature/transitional B cells in the bone marrow and spleen, energy demand seems to be relatively low [102]. In SLE, metabolic reprogramming directs the differentiation of unswitched memory B cells into plasmablasts or CD27- IgD- memory B cells; the former necessitates a shift of metabolism to anabolism with increased glycolysis and mechanistic target of rapamycin complex 1 (mTORC1) activation [103]. It would be interesting to determine how metabolic reprogramming regulates B cell proliferation, differentiation and function in IIMs.

Dendritic cells (DCs)

DCs are the most potent antigen-presenting cells and play a central role in initiating the immune response while maintaining

Table 1
The altered metabolites in IIMs.

Samples	Molecules	Clinical association	Reference
Glycolysis			
Muscle tissues			
DM	PKM2↑	a novel biomarker	[32]
IBM	GAPDH, LDHA, AGL, PFKM, FBP2, PGAM2↓	repression of glycolytic metabolism	[32,33]
FA metabolism			
Serum			
DM/PM	PA↑	correlated with atrophy of myotubes	[44]
	TG(46:1),TG(48:2), TG(48:1), TG(50:1) ↑	muscle inflammation and performance	[44]
DM/PM	AA↓	turns into pro-inflammatory chemicals	[44]
	PC(36:4), PC(38:5),PC(38:4), TG(48:5), TG (54:4) ↓	muscle inflammation and performance	[44]
PBMC			
PM	PA↑	a biomarker of PM	[45]
Muscle tissues			
IIM	Prostaglandins, leukotrienes↑	muscle atrophy, impairment, and pain	[48]
DM/PM	5-LO, FLAP, LTB4↑	negatively correlated with muscle performance	[47]
TCA cycle			
Muscle tissues			
DM/PM	SDH↑	abnormal muscle mitochondrial function	[61,62]
PM	CS, MDH, IDH↓	decreased TCA activity, abnormal muscle mitochondrial function	[61]
Amino acid metabolism			
serum/plasma			
IBM	Cystathionine, dimethylglycine, TCDCA, citrulline, alternative NAD + synthesis pathway, nucleotide synthesis precursors, Cr/ Cn ratio↑	respiratory chain deficiencies and depleted Cr pools	[67]
IBM	niacinamide↓	respiratory chain deficiencies	[67]
Urine			
DM/PM	Cr, taurine, glycine, citrate, and choline-containing compounds↑	depleted Cr pools	[68]

tolerance [104]. There are two main subtypes of DCs: myeloid DCs and pDCs. Myeloid DCs participate in the adaptive immune response and capture, process, and present antigens to prime T cells, while pDCs can produce large amounts of type I IFN [105]. Several studies have suggested that reductions in the numbers of blood DCs in autoimmune diseases seem to result from increased recruitment to target tissues [106]. Distinct DC subsets have been identified in IIM muscle tissue, suggesting an ongoing immune response. Both mature and immature DCs are present in muscle tissue in PM and DM, and the mature/immature DC ratio is higher in PM than in DM [107]. DCs in muscle from DM patients are mainly localized in perivascular infiltrates, whereas in PM, DCs seem to surround or invade muscle fibres [107,108]. In addition, pDCs are found mostly in DM muscle, while PM and IBM muscle has more myeloid DCs than pDCs. However, in PM patients with myositis-associated autoantibodies, especially anti-Jo-1 or anti-Ro/SSA antibodies, increased numbers of pDCs are observed in muscle tissue, which suggests a role for pDCs and type I IFN in antibody-positive PM [109]. PDCs are also observed in skin from patients with JDM with increased expression of type I IFN-induced proteins.

Upon activation by toll-like receptor (TLR) agonists, DCs undergo a metabolic shift towards glycolysis, which provides metabolic and bioenergetic material to support the function and survival of the cells [110–112]. Inhibition of hexokinase, the first key enzyme in the glycolysis pathway, prevents the whole DCs activation process, which highlights the importance of glucose for DC activation [113]. However, tolerogenic DCs favor FA oxidation [114]. Impaired FA production leads to decreased immunogenicity [115]. The balance between the immunogenicity and tolerogenicity of DCs reflects the balance between anabolism and catabolism. Moreover, DCs can sense and respond to multiple metabolic pathways and metabolites, such as the mammalian target of rapamycin (mTOR) pathway and extracellular succinate, respectively [116,117].

Macrophages

The proportion of macrophages in skeletal muscle differs in different clinical phenotypes of IIMs. Anti-HMGCR myopathy is defined as positive anti-HMGCR autoantibodies, elevated serum CK levels and proximal skeletal muscle weakness [118]. CD163 is a putative marker of M2 macrophages. Compared with DM patients, muscle biopsies from anti-HMGCR myopathy show that M2 macrophages are predominant over T cells, DCs and M1 macrophages [119]. Inflammatory myopathy with abundant macrophages may be a distinct IIM entity and is characterized by infiltration of muscle and connective tissue by numerous macrophages, among which M2 macrophages make up a majority [120,121]. Proteomic profiling and immunohistochemistry analysis of muscle biopsies from IBM patients suggest activated CD163+M2 macrophages in the pathogenesis of IBM [122]. The levels of soluble CD163 in the circulation are increased in DM/PM patients with ILD compared with those without ILD and healthy controls [123]. In IMNM, skeletal muscle tissue shows a Th1/M1 macrophage-mediated immune response [10]. It should be noted, however, that using cell surface markers to define M1 and M2 macrophages has some limitations. Myopathology in IIMs with anti-Jo-1 antibodies, as well as in juvenile IIMs with anti-NXP2 and anti-TIF1γ antibodies presents a macrophage predominant inflammation infiltration, but without classifying the macrophage subtypes [124,125]. Upon acute injury to skeletal muscle, M1 macrophages arrive first to play a role in clearing muscle necrotic debris, while M2 macrophages arrive later to participate in muscle repair and regeneration [126,127].

M1 macrophages utilize glycolysis and have a fragmented TCA cycle, which contributes to the accumulation of succinate and citrate [117,128–130]. In turn, citrate accumulation in M1 macrophages leads to the production of prostaglandins, ROS, and nitric oxide (NO). Succinate regulates NLRP3 inflammasome activity and induces IL-1β production [131,132]. Activation of the NLRP3 inflammasome may augment inflammation in IIMs. In contrast, M2 macrophages use FAs to support OXPHOS [129,133]. M2 polarization can activate UDP-GlcNAc-associated modules and glutamine catabolism [134].

Conclusion

Given the important role of metabolism in the regulation of muscle tissue and immune cells, fine-tuning key metabolic signaling pathways, metabolites, or metabolic enzymes may offer an attractive

alternative to immunosuppressive therapy for IIMs. Targeting of select metabolic processes has become an attractive treatment strategy for autoimmune diseases. For example, methotrexate, which can inhibit one-carbon metabolism, is the first-line treatment for RA [135]. Inhibiting the glycolytic kinase PDK1 can reduce Th17 cell-mediated inflammation in models of experimental autoimmune encephalomyelitis [136]. Combined treatment with the glucose metabolism inhibitor 2-deoxy-D-glucose (2-DG) and metformin can normalize CD4+ T cell metabolism, thereby reversing predisposition for lupus in mice [137]. Targeting glycolysis may inhibit activation of multiple immune cell types, such as effector T cells, and alleviate IIMs. Endurance exercise has been shown to activate aerobic metabolism in DM/PM patients while concomitantly suppressing the inflammatory response in muscles [18]. Due to the important role of FA metabolism in DM/PM, targeting the FA pathway is also a promising approach. Dietary supplementation with unsaturated FA has been shown to improve muscle function in DM/PM patients [56,57]. MTOR plays a central role in multiple metabolic programmes. Targeting mTOR signaling to regulate metabolism could contribute to immune suppression [138,139]. Mitochondria provides energy to sustain cell function and regulates metabolic reprogramming and immune signaling cascades [140,141]. Altered mitochondrial structure, genome, proteome and function occur in muscle tissues from DM, PM and IBM patients, suggesting a prominent role of mitochondrial energy-related metabolic pathways. Mitochondrial abnormalities are associated with the inhibition of TCA cycle and OXPHOS and excessive production of ROS, finally lead to oxidative damage, muscle inflammation and weakness [59,142,143]. Activation of pathways related mitochondrial biogenesis can correct the biochemical deficiency in animal models [144,145]. Rapamycin, the inhibitor of mTORC1, regulates mitochondrial stress response in mitochondrial diseases [146]. ROS scavengers show a role of relieving joint inflammation in an experimental model of osteoarthritis [147]. Intervention of mitochondrial dysfunction may be potential interventions for IIMs.

Over the past decade, with the development of high-throughput technologies, integrated information on metabolism has been collected that offers new clues into the role of metabolic reprogramming in IIMs. Most studies have focused on glycolysis and FA metabolism, providing valuable insights into the mechanisms of IIMs as well as novel treatment targets. In contrast, studies on the TCA cycle, the PPP and amino acid metabolism are scarce and are urgently needed. In addition to muscle cells, multiple types of immune cells participate in the pathogenesis of IIMs, and understanding of how immune cells undergo metabolic reprogramming and affect disease progression in IIMs is urgently needed. Nevertheless, the currently available studies have suggested that interventions in metabolic pathways could reduce muscle damage. We foresee that interference with metabolic checkpoints will be an attractive treatment strategy for IIMs and that previously unappreciated metabolic pathways or critical enzymes or metabolites will be found to be involved in the progression or suppression of IIMs, some of which may be candidate biomarkers.

Funding

This work was supported by National Natural Science Foundation of China (81771765, 81373206, 81671622), Hunan Provincial Natural Science Foundation (2018JJ3823) and Fundamental Research Funds for the Central Universities of Central South University (2018zzts247).

Author contributions

D.L. wrote first draft. X.Z. and H.L. revised the manuscript. H.Z. has revised final version and added extra information.

Ethical approval

NA (this is a review of literature and no patients or patient information was used in the writing of this article).

Declaration of Competing Interest

None.

Acknowledgments

None.

References

[1] Miller FW, Lamb JA, Schmidt J, Nagaraju K. Risk factors and disease mechanisms in myositis. Nat Rev Rheumatol 2018;14(5):255–68. doi: 10.1038/nrrheum.2018.48.
[2] Dimachkie MM, Barohn RJ, Amato AA. Idiopathic inflammatory myopathies. Neurol Clin 2014;32(3):595–628 vii. doi: 10.1016/j.ncl.2014.04.007.
[3] Rayavarapu S, Coley W, Kinder TB, Nagaraju K. Idiopathic inflammatory myopathies: pathogenic mechanisms of muscle weakness. Skelet Muscle 2013;3(1):13. doi: 10.1186/2044-5040-3-13.
[4] Svensson J, Arkema EV, Lundberg IE, Holmqvist M. Incidence and prevalence of idiopathic inflammatory myopathies in Sweden: a nationwide population-based study. Rheumatology 2017;56(5):802–10. doi: 10.1093/rheumatology/kew503.
[5] Meyer A, Meyer N, Schaeffer M, Gottenberg JE, Geny B, Sibilia J. Incidence and prevalence of inflammatory myopathies: a systematic review. Rheumatology 2015;54(1):50–63. doi: 10.1093/rheumatology/keu289.
[6] Zhong L, Yu Z, Song H. Association of anti-nuclear matrix protein 2 antibody with complications in patients with idiopathic inflammatory myopathies: a meta-analysis of 20 cohorts. Clin Immunol 2019;198:11–8. doi: 10.1016/j.clim.2018.11.008.
[7] Barba T, Fort R, Cottin V, Provencher S, Durieu I, Jardel S, et al. Treatment of idiopathic inflammatory myositis associated interstitial lung disease: a systematic review and meta-analysis. Autoimmun Rev 2019;18(2):113–22. doi: 10.1016/j.autrev.2018.07.013.
[8] Gherardi RK. Pathogenic aspects of dermatomyositis, polymyositis and overlap myositis. Presse Med 2011;40(4 Pt 2):e209–18. doi: 10.1016/j.lpm.2010.12.013.
[9] Dalakas MC. Sporadic inclusion body myositis–diagnosis, pathogenesis and therapeutic strategies. Nat Clin Pract Neurol 2006;2(8):437–47. doi: 10.1038/ncpneuro0261.
[10] Preusse C, Goebel HH, Held J, Wengert O, Scheibe F, Irlbacher K, et al. Immune-mediated necrotizing myopathy is characterized by a specific Th1-M1 polarized immune profile. Am J Pathol 2012;181(6):2161–71. doi: 10.1016/j.ajpath.2012.08.033.
[11] Reed AM, Ytterberg SR. Genetic and environmental risk factors for idiopathic inflammatory myopathies. Rheum Dis Clin N Am 2002;28(4):891–916.
[12] Day J, Otto S, Proudman S, Hayball JD, Limaye V. Dysregulated innate immune function in the aetiopathogenesis of idiopathic inflammatory myopathies. Autoimmun Rev 2017;16(1):87–95. doi: 10.1016/j.autrev.2016.09.019.
[13] Goebels N, Michaelis D, Engelhardt M, Huber S, Bender A, Pongratz D, et al. Differential expression of perforin in muscle-infiltrating T cells in polymyositis and dermatomyositis. J Clin Investig 1996;97(12):2905–10. doi: 10.1172/JCI118749.
[14] Pandya JM, Venalis P, Al-Khalili L, Shahadat Hossain M, Stache V, Lundberg IE, et al. CD4+ and CD8+ CD28(null) T cells are cytotoxic to autologous muscle cells in patients with polymyositis. Arthritis Rheumatol 2016;68(8):2016–26. doi: 10.1002/art.39650.
[15] Venalis P, Lundberg IE. Immune mechanisms in polymyositis and dermatomyositis and potential targets for therapy. Rheumatology 2014;53(3):397–405. doi: 10.1093/rheumatology/ket279.
[16] Orlandi M, Barsotti S, Cioffi E, Tenti S, Toscano C, Baldini C, et al. One year in review 2016: idiopathic inflammatory myopathies. Clin Exp Rheumatol 2016;34(6):966–74.
[17] Nader GA, Dastmalchi M, Alexanderson H, Grundtman C, Gernapudi R, Esbjornsson M, et al. A longitudinal, integrated, clinical, histological and mRNA profiling study of resistance exercise in myositis. Mol Med 2010;16(11–12):455–64. doi: 10.2119/molmed.2010.00016.
[18] Munters LA, Loell I, Ossipova E, Raouf J, Dastmalchi M, Lindroos E, et al. Endurance exercise improves molecular pathways of aerobic metabolism in patients with myositis. Arthritis Rheumatol 2016;68(7):1738–50. doi: 10.1002/art.39624.
[19] Coley W, Rayavarapu S, Nagaraju K. Role of non-immune mechanisms of muscle damage in idiopathic inflammatory myopathies. Arthritis Res Ther 2012;14(2):209. doi: 10.1186/ar3791.
[20] Wang A, Luan HH, Medzhitov R. An evolutionary perspective on immunometabolism. Science 2019;363(6423). doi: 10.1126/science.aar3932.
[21] Zhu H, Chen W, Liu D, Luo H. The role of metabolism in the pathogenesis of systemic sclerosis. Metab Clin Exp 2019;93:44–51. doi: 10.1016/j.metabol.2018.12.004.
[22] O'Neill LA, Kishton RJ, Rathmell J. A guide to immunometabolism for immunologists. Nat Rev Immunol 2016;16(9):553–65. doi: 10.1038/nri.2016.70.

[23] Singer BD, Chandel NS. Immunometabolism of pro-repair cells. J Clin Investig 2019;130:2597–607. doi: 10.1172/JCI124613.
[24] Lightfoot YL, Blanco LP, Kaplan MJ. Metabolic abnormalities and oxidative stress in lupus. Curr Opin Rheumatol 2017;29(5):442–9. doi: 10.1097/BOR.0000000000000413.
[25] Freitag J, Berod L, Kamradt T, Sparwasser T. Immunometabolism and autoimmunity. Immunol Cell Biol 2016;94(10):925–34. doi: 10.1038/icb.2016.77.
[26] Rhoads JP, Major AS, Rathmell JC. Fine tuning of immunometabolism for the treatment of rheumatic diseases. Nat Rev Rheumatol 2017;13(5):313–20. doi: 10.1038/nrrheum.2017.54.
[27] Wahl DR, Petersen B, Warner R, Richardson BC, Glick GD, Opipari AW. Characterization of the metabolic phenotype of chronically activated lymphocytes. Lupus 2010;19(13):1492–501. doi: 10.1177/0961203310373109.
[28] Weyand CM, Zeisbrich M, Goronzy JJ. Metabolic signatures of T-cells and macrophages in rheumatoid arthritis. Curr Opin Immunol 2017;46:112–20. doi: 10.1016/j.coi.2017.04.010.
[29] Garcia-Carbonell R, Divakaruni AS, Lodi A, Vicente-Suarez I, Saha A, Cheroutre H, et al. Critical role of glucose metabolism in rheumatoid arthritis fibroblast-like synoviocytes. Arthritis Rheumatol 2016;68(7):1614–26. doi: 10.1002/art.39608.
[30] Wu CL, Satomi Y, Walsh K. RNA-seq and metabolomic analyses of Akt1-mediated muscle growth reveals regulation of regenerative pathways and changes in the muscle secretome. BMC Genomics 2017;18(1):181. doi: 10.1186/s12864-017-3548-2.
[31] Li Y, Zhou Y, Wang Q. Multiple values of (18)F-FDG PET/CT in idiopathic inflammatory myopathy. Clin Rheumatol 2017;36(10):2297–305. doi: 10.1007/s10067-017-3794-3.
[32] Santacatterina F, Sanchez-Arago M, Catalan-Garcia M, Garrabou G, de Arenas CN, Grau JM, et al. Pyruvate kinase M2 and the mitochondrial ATPase Inhibitory Factor 1 provide novel biomarkers of dermatomyositis: a metabolic link to oncogenesis. J Transl Med 2017;15(1):29. doi: 10.1186/s12967-017-1136-5.
[33] Parker KC, Kong SW, Walsh RJ, Salajegheh M, Moghadaszadeh B, Amato AA, et al. Fast-twitch sarcomeric and glycolytic enzyme protein loss in inclusion body myositis. Muscle Nerve 2009;39(6):739–53. doi: 10.1002/mus.21230.
[34] Xiao Y, Zhu H, Li L, Gao S, Liu D, Dai B, et al. Global analysis of protein expression in muscle tissues of dermatomyositis/polymyosisits patients demonstrated an association between dysferlin and human leucocyte antigen A. Rheumatology 2019. doi: 10.1093/rheumatology/kez085.
[35] Rayavarapu S, Coley W, Van der Meulen JH, Cakir E, Tappeta K, Kinder TB, et al. Activation of the ubiquitin proteasome pathway in a mouse model of inflammatory myopathy: a potential therapeutic target. Arthritis Rheum 2013;65(12):3248–58. doi: 10.1002/art.38180.
[36] Coley W, Rayavarapu S, Pandey GS, Sabina RL, Van der Meulen JH, Ampong B, et al. The molecular basis of skeletal muscle weakness in a mouse model of inflammatory myopathy. Arthritis Rheum 2012;64(11):3750–9. doi: 10.1002/art.34625.
[37] Mammen AL. Statin-associated autoimmune myopathy. N Engl J Med 2016;374(7):664–9. doi: 10.1056/NEJMra1515161.
[38] Tiniakou E, Christopher-Stine L. Immune-mediated necrotizing myopathy associated with statins: history and recent developments. Curr Opin Rheumatol 2017;29(6):604–11. doi: 10.1097/BOR.0000000000000438.
[39] Taylor BA, Thompson PD. Muscle-related side-effects of statins: from mechanisms to evidence-based solutions. Curr Opin Lipidol 2015;26(3):221–7. doi: 10.1097/MOL.0000000000000174.
[40] de Moraes MT, de Souza FH, de Barros TB, Shinjo SK. Analysis of metabolic syndrome in adult dermatomyositis with a focus on cardiovascular disease. Arthritis Care Res 2013;65(5):793–9. doi: 10.1002/acr.21879.
[41] Araujo PAO, Silva MG, Borba EF, Shinjo SK. High prevalence of metabolic syndrome in antisynthetase syndrome. Clin Exp Rheumatol 2018;36(2):241–7.
[42] Wang H, Tang J, Chen X, Li F, Luo J. Lipid profiles in untreated patients with dermatomyositis. J Eur Acad Dermatol Venereol 2013;27(2):175–9 http://doi.org/10.1111/j.1468-3083.2011.04437.x.
[43] Wang H, Cai Y, Cai L, Hu Y, Chen X, Deng J. Altered lipid levels in untreated patients with early polymyositis. PLoS ONE 2014;9(2):e89827. doi: 10.1371/journal.pone.0089827.
[44] Raouf J, Idborg H, Englund P, Alexanderson H, Dastmalchi M, Jakobsson PJ, et al. Targeted lipidomics analysis identified altered serum lipid profiles in patients with polymyositis and dermatomyositis. Arthritis Res Ther 2018;20(1):83. doi: 10.1186/s13075-018-1579-y.
[45] Yin G, Wang Y, Cen XM, Yang Y, Yang M, Xie QB. Identification of palmitoleic acid controlled by mtor signaling as a biomarker of polymyositis. J Immunol Res 2017;2017:3262384. doi: 10.1155/2017/3262384.
[46] Korotkova M, Lundberg IE. The skeletal muscle arachidonic acid cascade in health and inflammatory disease. Nat Rev Rheumatol 2014;10(5):295–303. doi: 10.1038/nrrheum.2014.2.
[47] Loell I, Alemo Munters L, Pandya J, Zong M, Alexanderson H, Fasth AE, et al. Activated LTB4 pathway in muscle tissue of patients with polymyositis or dermatomyositis. Ann Rheum Dis 2013;72(2):293–9. doi: 10.1136/annrheumdis-2012-201294.
[48] Korotkova M, Helmers SB, Loell I, Alexanderson H, Grundtman C, Dorph C, et al. Effects of immunosuppressive treatment on microsomal prostaglandin E synthase 1 and cyclooxygenases expression in muscle tissue of patients with polymyositis or dermatomyositis. Ann Rheum Dis 2008;67(11):1596–602. doi: 10.1136/ard.2007.079525.
[49] Bryner RW, Woodworth-Hobbs ME, Williamson DL, Alway SE. Docosahexaenoic acid protects muscle cells from palmitate-induced atrophy. ISRN Obes 2012;2012:647348. doi: 10.5402/2012/647348.
[50] Lipina C, Hundal HS. Lipid modulation of skeletal muscle mass and function. J Cachexia Sarcopenia Muscle 2017;8(2):190–201. doi: 10.1002/jcsm.12144.
[51] Loell I, Raouf J, Chen YW, Shi R, Nennesmo I, Alexanderson H, et al. Effects on muscle tissue remodeling and lipid metabolism in muscle tissue from adult patients with polymyositis or dermatomyositis treated with immunosuppressive agents. Arthritis Res Ther 2016;18(1):136. doi: 10.1186/s13075-016-1033-y.
[52] Huang YS, Huang WC, Li CW, Chuang LT. Eicosadienoic acid differentially modulates production of pro-inflammatory modulators in murine macrophages. Mol Cell Biochem 2011;358(1–2):85–94. doi: 10.1007/s11010-011-0924-0.
[53] Vasiljevski ER, Summers MA, Little DG, Schindeler A. Lipid storage myopathies: current treatments and future directions. Prog Lipid Res 2018;72:1–17. doi: 10.1016/j.plipres.2018.08.001.
[54] Jaworska-Wilczynska M, Wilczynski GM, Engel WK, Strickland DK, Weisgraber KH, Askanas V. Three lipoprotein receptors and cholesterol in inclusion-body myositis muscle. Neurology 2002;58(3):438–45. doi: 10.1212/wnl.58.3.438.
[55] Verma S, Singh S, Bhalla AK, Khullar M. Study of subcutaneous fat in children with juvenile dermatomyositis. Arthritis Rheum 2006;55(4):564–8. doi: 10.1002/art.22108.
[56] Smith GI, Julliand S, Reeds DN, Sinacore DR, Klein S, Mittendorfer B. Fish oil-derived n-3 PUFA therapy increases muscle mass and function in healthy older adults. Am J Clin Nutr 2015;102(1):115–22. doi: 10.3945/ajcn.114.105833.
[57] Rodacki CL, Rodacki AL, Pereira G, Naliwaiko K, Coelho I, Pequito D, et al. Fish-oil supplementation enhances the effects of strength training in elderly women. Am J Clin Nutr 2012;95(2):428–36. doi: 10.3945/ajcn.111.021915.
[58] Oldfors A, Moslemi AR, Jonasson L, Ohlsson M, Kollberg G, Lindberg C. Mitochondrial abnormalities in inclusion-body myositis. Neurology 2006;66(2 Suppl 1):S49–55. doi: 10.1212/01.wnl.0000192127.63013.8d.
[59] Joshi PR, Vetterke M, Hauburger A, Tacik P, Stoltenburg G, Hanisch F. Functional relevance of mitochondrial abnormalities in sporadic inclusion body myositis. J Clin Neurosci: Off J Neurosurg SocAustralas 2014;21(11):1959–63. doi: 10.1016/j.jocn.2014.05.051.
[60] Katsetos CD, Koutzaki S, Melvin JJ. Mitochondrial dysfunction in neuromuscular disorders. Semin Pediatr Neurol 2013;20(3):202–15. doi: 10.1016/j.spen.2013.10.010.
[61] Sunitha B, Gayathri N, Kumar M, Keshava Prasad TS, Nalini A, Padmanabhan B, et al. Muscle biopsies from human muscle diseases with myopathic pathology reveal common alterations in mitochondrial function. J Neurochem 2016;138(1):174–91. doi: 10.1111/jnc.13626.
[62] Meyer A, Laverny G, Allenbach Y, Grelet E, Ueberschlag V, Echaniz-Laguna A, et al. IFN-beta-induced reactive oxygen species and mitochondrial damage contribute to muscle impairment and inflammation maintenance in dermatomyositis. Acta Neuropathol 2017;134(4):655–66. doi: 10.1007/s00401-017-1731-9.
[63] Alhatou MI, Sladky JT, Bagasra O, Glass JD. Mitochondrial abnormalities in dermatomyositis: characteristic pattern of neuropathology. J Mol Histol 2004;35(6):615–9. doi: 10.1007/s10735-004-2194-6.
[64] Boncompagni S, Moussa CE, Levy E, Pezone MJ, Lopez JR, Protasi F, et al. Mitochondrial dysfunction in skeletal muscle of amyloid precursor protein-overexpressing mice. J Biol Chem 2012;287(24):20534–44. doi: 10.1074/jbc.M112.359588.
[65] Varadhachary AS, Weihl CC, Pestronk A. Mitochondrial pathology in immune and inflammatory myopathies. Curr Opin Rheumatol 2010;22(6):651–7. doi: 10.1097/BOR.0b013e32833f108a.
[66] Lightfoot AP, Nagaraju K, McArdle A, Cooper RG. Understanding the origin of non-immune cell-mediated weakness in the idiopathic inflammatory myopathies - potential role of ER stress pathways. Curr Opin Rheumatol 2015;27(6):580–5. doi: 10.1097/BOR.0000000000000212.
[67] Buzkova J, Nikkanen J, Ahola S, Hakonen AH, Sevastianova K, Hovinen T, et al. Metabolomes of mitochondrial diseases and inclusion body myositis patients: treatment targets and biomarkers. EMBO Mol Med 2018;10(12). doi: 10.15252/emmm.201809091.
[68] Chung YL, Wassif WS, Bell JD, Hurley M, Scott DL. Urinary levels of creatine and other metabolites in the assessment of polymyositis and dermatomyositis. Rheumatology 2003;42(2):298–303. doi: 10.1093/rheumatology/keg084.
[69] Chung YL, Rider LG, Bell JD, Summers RM, Zemel LS, Rennebohm RM, et al. Muscle metabolites, detected in urine by proton spectroscopy, correlate with disease damage in juvenile idiopathic inflammatory myopathies. Arthritis Rheum 2005;53(4):565–70. doi: 10.1002/art.21331.
[70] Barbieri E, Guescini M, Calcabrini C, Vallorani L, Diaz AR, Fimognari C, et al. Creatine prevents the structural and functional damage to mitochondria in myogenic, oxidatively stressed C2C12 cells and restores their differentiation capacity. Oxid Med Cell Longev 2016;2016:5152029. doi: 10.1155/2016/5152029.
[71] Chung YL, Alexanderson H, Pipitone N, Morrison C, Dastmalchi M, Stahl-Hallengren C, et al. Creatine supplements in patients with idiopathic inflammatory myopathies who are clinically weak after conventional pharmacologic treatment: six-month, double-blind, randomized, placebo-controlled trial. Arthritis Rheum 2007;57(4):694–702. doi: 10.1002/art.22687.
[72] Buck MD, Sowell RT, Kaech SM, Pearce EL. Metabolic instruction of immunity. Cell 2017;169(4):570–86. doi: 10.1016/j.cell.2017.04.004.
[73] Wang DX, Lu X, Zu N, Lin B, Wang LY, Shu XM, et al. Clinical significance of peripheral blood lymphocyte subsets in patients with polymyositis and dermatomyositis. Clin Rheumatol 2012;31(12):1691–7. doi: 10.1007/s10067-012-2075-4.
[74] Espinosa-Ortega F, Gomez-Martin D, Santana-De Anda K, Romo-Tena J, Villasenor-Ovies P, Alcocer-Varela J. Quantitative T cell subsets profile in peripheral blood from patients with idiopathic inflammatory myopathies: tilting the

balance towards proinflammatory and pro-apoptotic subsets. Clin Exp Immunol 2015;179(3):520–8. doi: 10.1111/cei.12475.
[75] Tang X, Tian X, Zhang Y, Wu W, Tian J, Rui K, et al. Correlation between the frequency of Th17 cell and the expression of microRNA-206 in patients with dermatomyositis. Clin Dev Immunol 2013;2013:345347. doi: 10.1155/2013/345347.
[76] Banica L, Besliu A, Pistol G, Stavaru C, Ionescu R, Forsea AM, et al. Quantification and molecular characterization of regulatory T cells in connective tissue diseases. Autoimmunity 2009;42(1):41–9. doi: 10.1080/08916930802282651.
[77] Orimo S, Koga R, Goto K, Nakamura K, Arai M, Tamaki M, et al. Immunohistochemical analysis of perforin and granzyme A in inflammatory myopathies. Neuromusc Disord 1994;4(3):219–26.
[78] Fasth AE, Dastmalchi M, Rahbar A, Salomonsson S, Pandya JM, Lindroos E, et al. T cell infiltrates in the muscles of patients with dermatomyositis and polymyositis are dominated by CD28null T cells. J Immunol 2009;183(7):4792–9. doi: 10.4049/jimmunol.0803688.
[79] Pandya JM, Fasth AE, Zong M, Arnardottir S, Dani L, Lindroos E, et al. Expanded T cell receptor Vbeta-restricted T cells from patients with sporadic inclusion body myositis are proinflammatory and cytotoxic CD28null T cells. Arthritis Rheum 2010;62(11):3457–66. doi: 10.1002/art.27665.
[80] Waschbisch A, Schwab N, Ruck T, Stenner MP, Wiendl H. FOXP3+ T regulatory cells in idiopathic inflammatory myopathies. J Neuroimmunol 2010;225 (1–2):137–42. doi: 10.1016/j.jneuroim.2010.03.013.
[81] Pearce EL, Walsh MC, Cejas PJ, Harms GM, Shen H, Wang LS, et al. Enhancing CD8 T-cell memory by modulating fatty acid metabolism. Nature 2009;460 (7251):103–7. doi: 10.1038/nature08097.
[82] Chang CH, Pearce EL. Emerging concepts of T cell metabolism as a target of immunotherapy. Nat Immunol 2016;17(4):364–8. doi: 10.1038/ni.3415.
[83] Patel CH, Powell JD. Targeting T cell metabolism to regulate T cell activation, differentiation and function in disease. Curr Opin Immunol 2017;46:82–8. doi: 10.1016/j.coi.2017.04.006.
[84] Michalek RD, Gerriets VA, Jacobs SR, Macintyre AN, MacIver NJ, Mason EF, et al. Cutting edge: distinct glycolytic and lipid oxidative metabolic programs are essential for effector and regulatory CD4+ T cell subsets. J Immunol 2011;186 (6):3299–303. doi: 10.4049/jimmunol.1003613.
[85] Barbi J, Pardoll D, Pan F. Metabolic control of the Treg/Th17 axis. Immunol Rev 2013;252(1):52–77. doi: 10.1111/imr.12029.
[86] Greenberg SA, Bradshaw EM, Pinkus JL, Pinkus GS, Burleson T, Due B, et al. Plasma cells in muscle in inclusion body myositis and polymyositis. Neurology 2005;65(11):1782–7. doi: 10.1212/01.wnl.0000187124.92826.20.
[87] Ishii W, Matsuda M, Shimojima Y, Itoh S, Sumida T, Ikeda S. Flow cytometric analysis of lymphocyte subpopulations and TH1/TH2 balance in patients with polymyositis and dermatomyositis. Intern Med 2008;47(18):1593–9. doi: 10.2169/internalmedicine.47.0967.
[88] Piper CJM, Wilkinson MGL, Deakin CT, Otto GW, Dowle S, Duurland CL, et al. CD19(+)CD24(hi)CD38(hi) B Cells are expanded in juvenile dermatomyositis and exhibit a pro-inflammatory phenotype after activation through toll-like receptor 7 and interferon-alpha. Front Immunol 2018;9:1372. doi: 10.3389/fimmu.2018.01372.
[89] Li W, Tian X, Lu X, Peng Q, Shu X, Yang H, et al. Significant decrease in peripheral regulatory B cells is an immunopathogenic feature of dermatomyositis. Sci Rep 2016;6:27479. doi: 10.1038/srep27479.
[90] Tansley SL, McHugh NJ. Myositis specific and associated autoantibodies in the diagnosis and management of juvenile and adult idiopathic inflammatory myopathies. Curr Rheumatol Rep 2014;16(12):464. doi: 10.1007/s11926-014-0464-1.
[91] Wedderburn LR, Varsani H, Li CK, Newton KR, Amato AA, Banwell B, et al. International consensus on a proposed score system for muscle biopsy evaluation in patients with juvenile dermatomyositis: a tool for potential use in clinical trials. Arthritis Rheum 2007;57(7):1192–201. doi: 10.1002/art.23012.
[92] Varsani H, Charman SC, Li CK, Marie SK, Amato AA, Banwell B, et al. Validation of a score tool for measurement of histological severity in juvenile dermatomyositis and association with clinical severity of disease. Ann Rheum Dis 2015;74 (1):204–10. doi: 10.1136/annrheumdis-2013-203396.
[93] Betteridge Z, McHugh N. Myositis-specific autoantibodies: an important tool to support diagnosis of myositis. J Intern Med 2016;280(1):8–23. doi: 10.1111/joim.12451.
[94] Krystufkova O, Vallerskog T, Helmers SB, Mann H, Putova I, Belacek J, et al. Increased serum levels of B cell activating factor (BAFF) in subsets of patients with idiopathic inflammatory myopathies. Ann Rheum Dis 2009;68(6):836–43. doi: 10.1136/ard.2008.091405.
[95] Peng QL, Xie Y, Zhang RF, Shu XM, Wang GC. [The levels and clinical significance of serum B cell activating factor in Chinese patients with polymyositis or dermatomyositis]. Zhonghua Nei Ke Za Zhi 2012;51(3):210–3.
[96] Musset L, Allenbach Y, Benveniste O, Boyer O, Bossuyt X, Bentow C, et al. Anti-HMGCR antibodies as a biomarker for immune-mediated necrotizing myopathies: a history of statins and experience from a large international multi-center study. Autoimmun Rev 2016;15(10):983–93. doi: 10.1016/j.autrev.2016.07.023.
[97] Jellusova J, Cato MH, Apgar JR, Ramezani-Rad P, Leung CR, Chen C, et al. Gsk3 is a metabolic checkpoint regulator in B cells. Nat Immunol 2017;18(3):303–12. doi: 10.1038/ni.3664.
[98] Doughty CA, Bleiman BF, Wagner DJ, Dufort FJ, Mataraza JM, Roberts MF, et al. Antigen receptor-mediated changes in glucose metabolism in B lymphocytes: role of phosphatidylinositol 3-kinase signaling in the glycolytic control of growth. Blood 2006;107(11):4458–65. doi: 10.1182/blood-2005-12-4788.
[99] Woodland RT, Fox CJ, Schmidt MR, Hammerman PS, Opferman JT, Korsmeyer SJ, et al. Multiple signaling pathways promote B lymphocyte stimulator dependent B-cell growth and survival. Blood 2008;111(2):750–60. doi: 10.1182/blood-2007-03-077222.
[100] Jellusova J, Rickert RC. The PI3K pathway in B cell metabolism. Crit Rev Biochem Mol Biol 2016;51(5):359–78. doi: 10.1080/10409238.2016.1215288.
[101] Stein M, Dutting S, Mougiakakos D, Bosl M, Fritsch K, Reimer D, et al. A defined metabolic state in pre B cells governs B-cell development and is counterbalanced by Swiprosin-2/EFhd1. Cell Death Differ 2017;24(7):1239–52. doi: 10.1038/cdd.2017.52.
[102] Boothby M, Rickert RC. Metabolic regulation of the immune humoral response. Immunity 2017;46(5):743–55. doi: 10.1016/j.immuni.2017.04.009.
[103] Torigoe M, Iwata S, Nakayamada S, Sakata K, Zhang M, Hajime M, et al. Metabolic reprogramming commits differentiation of human CD27(+)IgD(+) B cells to plasmablasts or CD27(-)IgD(-) cells. J Immunol 2017;199(2):425–34. doi: 10.4049/jimmunol.1601908.
[104] Dacka E, Bis B. Primary segmental infarction of the great omentum. Pol Przegl Chir 1976;48(6):721–3.
[105] Grundtman C, Malmstrom V, Lundberg IE. Immune mechanisms in the pathogenesis of idiopathic inflammatory myopathies. Arthritis Res Ther 2007;9 (2):208. doi: 10.1186/ar2139.
[106] Coutant F, Miossec P. Altered dendritic cell functions in autoimmune diseases: distinct and overlapping profiles. Nat Rev Rheumatol 2016;12(12):703–15. doi: 10.1038/nrrheum.2016.147.
[107] Page G, Chevrel G, Miossec P. Anatomic localization of immature and mature dendritic cell subsets in dermatomyositis and polymyositis: interaction with chemokines and Th1 cytokine-producing cells. Arthritis Rheum 2004;50 (1):199–208. doi: 10.1002/art.11428.
[108] Lopez de Padilla CM, Vallejo AN, McNallan KT, Vehe R, Smith SA, Dietz AB, et al. Plasmacytoid dendritic cells in inflamed muscle of patients with juvenile dermatomyositis. Arthritis Rheum 2007;56(5):1658–68. doi: 10.1002/art.22558.
[109] Eloranta ML, Barbasso Helmers S, Ulfgren AK, Ronnblom L, Alm GV, Lundberg IE. A possible mechanism for endogenous activation of the type I interferon system in myositis patients with anti-Jo-1 or anti-Ro 52/anti-Ro 60 autoantibodies. Arthritis Rheum 2007;56(9):3112–24. doi: 10.1002/art.22860.
[110] Krawczyk CM, Holowka T, Sun J, Blagih J, Amiel E, DeBerardinis RJ, et al. Toll-like receptor-induced changes in glycolytic metabolism regulate dendritic cell activation. Blood 2010;115(23):4742–9. doi: 10.1182/blood-2009-10-249540.
[111] Jantsch J, Chakravortty D, Turza N, Prechtel AT, Buchholz B, Gerlach RG, et al. Hypoxia and hypoxia-inducible factor-1 alpha modulate lipopolysaccharide-induced dendritic cell activation and function. J Immunol 2008;180(7):4697–705. doi: 10.4049/jimmunol.180.7.4697.
[112] Guak H, Al Habyan S, Ma EH, Aldossary H, Al-Masri M, Won SY, et al. Glycolytic metabolism is essential for CCR7 oligomerization and dendritic cell migration. Nat Commun 2018;9(1):2463. doi: 10.1038/s41467-018-04804-6.
[113] Everts B, Amiel E, Huang SC, Smith AM, Chang CH, Lam WY, et al. TLR-driven early glycolytic reprogramming via the kinases TBK1-IKKvarepsilon supports the anabolic demands of dendritic cell activation. Nat Immunol 2014;15 (4):323–32. doi: 10.1038/ni.2833.
[114] Sim WJ, Ahl PJ, Connolly JE. Metabolism is central to tolerogenic dendritic cell function. Mediators Inflamm 2016;2016:2636701. doi: 10.1155/2016/2636701.
[115] de Lima Thomaz L, Peron G, Oliveira J, da Rosa LC, Thome R, Verinaud L. The impact of metabolic reprogramming on dendritic cell function. Int Immunopharmacol 2018;63:84–93. doi: 10.1016/j.intimp.2018.07.031.
[116] Rubic T, Lametschwandtner G, Jost S, Hinteregger S, Kund J, Carballido-Perrig N, et al. Triggering the succinate receptor GPR91 on dendritic cells enhances immunity. Nat Immunol 2008;9(11):1261–9. doi: 10.1038/ni.1657.
[117] O'Neill LA, Pearce EJ. Immunometabolism governs dendritic cell and macrophage function. J Exp Med 2016;213(1):15–23. doi: 10.1084/jem.20151570.
[118] Allenbach Y, Mammen AL, Benveniste O, Stenzel W. 224th ENMC international workshop: clinico-sero-pathological classification of immune-mediated necrotizing myopathies Zandvoort, The Netherlands, 14-16 October 2016. Neuromusc Disord 2018;28(1):87–99. doi: 10.1016/j.nmd.2017.09.016.
[119] Chung T, Christopher-Stine L, Paik JJ, Corse A, Mammen AL. The composition of cellular infiltrates in anti-HMG-CoA reductase-associated myopathy. Muscle Nerve 2015;52(2):189–95. doi: 10.1002/mus.24642.
[120] Rinnenthal JL, Goebel HH, Preusse C, Lebenheim L, Schumann M, Moos V, et al. Inflammatory myopathy with abundant macrophages (IMAM): the immunology revisited. Neuromusc Disord 2014;24(2):151–5. doi: 10.1016/j.nmd.2013.11.004.
[121] Bassez G, Authier FJ, Lechapt-Zalcman E, Delfau-Larue MH, Plonquet A, Coquet M, et al. Inflammatory myopathy with abundant macrophages (IMAM): a condition sharing similarities with cytophagic histiocytic panniculitis and distinct from macrophagic myofasciitis. J Neuropathol Exp Neurol 2003;62(5):464–74. doi: 10.1093/jnen/62.5.464.
[122] Roos A, Preusse C, Hathazi D, Goebel HH, Stenzel W. Proteomic profiling unravels a key role of specific macrophage subtypes in sporadic inclusion body myositis. Front Immunol 2019;10:1040. doi: 10.3389/fimmu.2019.01040.
[123] Enomoto Y, Suzuki Y, Hozumi H, Mori K, Kono M, Karayama M, et al. Clinical significance of soluble CD163 in polymyositis-related or dermatomyositis-related interstitial lung disease. Arthritis Res Ther 2017;19(1):9. doi: 10.1186/s13075-016-1214-8.
[124] Mozaffar T, Pestronk A. Myopathy with anti-Jo-1 antibodies: pathology in perimysium and neighbouring muscle fibres. J Neurol Neurosurg Psychiatr 2000;68 (4):472–8. doi: 10.1136/jnnp.68.4.472.
[125] Yasin SA, Schutz PW, Deakin CT, Sag E, Varsani H, Simou S, et al. Histological heterogeneity in a large clinical cohort of juvenile idiopathic inflammatory myopathy: analysis by myositis autoantibody and pathological features. Neuropathol Appl Neurobiol 2019;45(5):495–512. doi: 10.1111/nan.12528.

[126] Liu X, Liu Y, Zhao L, Zeng Z, Xiao W, Chen P. Macrophage depletion impairs skeletal muscle regeneration: the roles of regulatory factors for muscle regeneration. Cell Biol Int 2017;41(3):228–38. doi: 10.1002/cbin.10705.
[127] Rigamonti E, Zordan P, Sciorati C, Rovere-Querini P, Brunelli S. Macrophage plasticity in skeletal muscle repair. Biomed Res Int 2014;2014:560629. doi: 10.1155/2014/560629.
[128] Odegaard JI, Chawla A. Alternative macrophage activation and metabolism. Annu Rev Pathol 2011;6:275–97. doi: 10.1146/annurev-pathol-011110-130138.
[129] Pearce EL, Pearce EJ. Metabolic pathways in immune cell activation and quiescence. Immunity 2013;38(4):633–43. doi: 10.1016/j.immuni.2013.04.005.
[130] Galvan-Pena S, O'Neill LA. Metabolic reprograming in macrophage polarization. Front Immunol 2014;5:420. doi: 10.3389/fimmu.2014.00420.
[131] Infantino V, Convertini P, Cucci L, Panaro MA, Di Noia MA, Calvello R, et al. The mitochondrial citrate carrier: a new player in inflammation. Biochem J 2011;438(3):433–6. doi: 10.1042/BJ20111275.
[132] Tannahill GM, Curtis AM, Adamik J, Palsson-McDermott EM, McGettrick AF, Goel G, et al. Succinate is an inflammatory signal that induces IL-1beta through HIF-1alpha. Nature 2013;496(7444):238–42. doi: 10.1038/nature11986.
[133] Vats D, Mukundan L, Odegaard JI, Zhang L, Smith KL, Morel CR, et al. Oxidative metabolism and PGC-1beta attenuate macrophage-mediated inflammation. Cell Metab 2006;4(1):13–24. doi: 10.1016/j.cmet.2006.05.011.
[134] Jha AK, Huang SC, Sergushichev A, Lampropoulou V, Ivanova Y, Loginicheva E, et al. Network integration of parallel metabolic and transcriptional data reveals metabolic modules that regulate macrophage polarization. Immunity 2015;42(3):419–30. doi: 10.1016/j.immuni.2015.02.005.
[135] Shuvalov O, Petukhov A, Daks A, Fedorova O, Vasileva E, Barlev NA. One-carbon metabolism and nucleotide biosynthesis as attractive targets for anticancer therapy. Oncotarget 2017;8(14):23955–77. doi: 10.18632/oncotarget.15053.
[136] Gerriets VA, Kishton RJ, Nichols AG, Macintyre AN, Inoue M, Ilkayeva O, et al. Metabolic programming and PDHK1 control CD4+ T cell subsets and inflammation. J Clin Investig 2015;125(1):194–207. doi: 10.1172/JCI76012.
[137] Yin Y, Choi SC, Xu Z, Perry DJ, Seay H, Croker BP, et al. Normalization of CD4+ T cell metabolism reverses lupus. Sci Transl Med 2015;7(274):274ra18. doi: 10.1126/scitranslmed.aaa0835.
[138] Waickman AT, Powell JD. mTOR, metabolism, and the regulation of T-cell differentiation and function. Immunol Rev 2012;249(1):43–58. doi: 10.1111/j.1600-065X.2012.01152.x.
[139] Sukhbaatar N, Hengstschlager M, Weichhart T. mTOR-mediated regulation of dendritic cell differentiation and function. Trends Immunol. 2016;37(11):778–89. doi: 10.1016/j.it.2016.08.009.
[140] Shutt TE, Shadel GS. A compendium of human mitochondrial gene expression machinery with links to disease. Environ Mol Mutagen 2010;51(5):360–79. doi: 10.1002/em.20571.
[141] Monlun M, Hyernard C, Blanco P, Lartigue L, Faustin B. Mitochondria as molecular platforms integrating multiple innate immune signalings. J Mol Biol 2017;429(1):1–13. doi: 10.1016/j.jmb.2016.10.028.
[142] Catalan-Garcia M, Garrabou G, Moren C, Guitart-Mampel M, Hernando A, Diaz-Ramos A, et al. Mitochondrial DNA disturbances and deregulated expression of oxidative phosphorylation and mitochondrial fusion proteins in sporadic inclusion body myositis. Clin Sci 2016;130(19):1741–51. doi: 10.1042/CS20160080.
[143] Mobasheri A, Rayman MP, Gualillo O, Sellam J, van der Kraan P, Fearon U. The role of metabolism in the pathogenesis of osteoarthritis. Nat Revi Rheumatol 2017;13(5):302–11. doi: 10.1038/nrrheum.2017.50.
[144] Viscomi C, Bottani E, Civiletto G, Cerutti R, Moggio M, Fagiolari G, et al. In vivo correction of COX deficiency by activation of the AMPK/PGC-1alpha axis. Cell Metab 2011;14(1):80–90. doi: 10.1016/j.cmet.2011.04.011.
[145] Cerutti R, Pirinen E, Lamperti C, Marchet S, Sauve AA, Li W, et al. NAD(+)-dependent activation of Sirt1 corrects the phenotype in a mouse model of mitochondrial disease. Cell Metab 2014;19(6):1042–9. doi: 10.1016/j.cmet.2014.04.001.
[146] Khan NA, Nikkanen J, Yatsuga S, Jackson C, Wang L, Pradhan S, et al. mTORC1 regulates mitochondrial integrated stress response and mitochondrial myopathy progression. Cell Metab 2017;26(2):419–28 e5. doi: 10.1016/j.cmet.2017.07.007.
[147] Nakagawa S, Arai Y, Mazda O, Kishida T, Takahashi KA, Sakao K, et al. N-acetylcysteine prevents nitric oxide-induced chondrocyte apoptosis and cartilage degeneration in an experimental model of osteoarthritis. J Orthopaed Res: Offi Publ Orthopaed Res Soc 2010;28(2):156–63. doi: 10.1002/jor.20976.